



AmelioSense™

Calm the flame of pyroptosis – soothe irritated skin

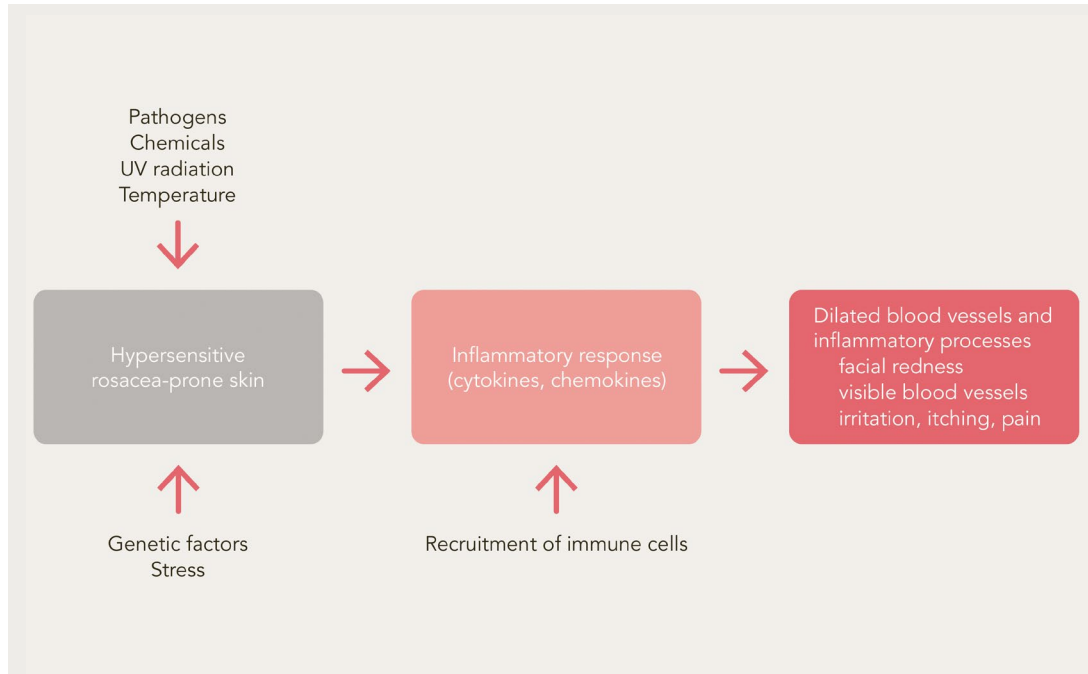
Rosacea

A Complex Inflammatory Process



- Rosacea is a complex **chronic inflammatory skin condition**.
- It primarily affects the face, especially the cheek and chin areas.
- Symptoms: **redness** and visible **blood vessels**, **itching**, **flushing**, and **pain**.
- Significant reduction in **quality of life**, self-esteem, and general **wellbeing**.
- Treatment of rosacea is challenging → topical anti-inflammatories, antibiotics, and sunscreens.

The Process of Rosacea in Hypersensitive Skin



Rosacea can be triggered by numerous factors such as chemical stimuli, environmental factors, pathogens, and genetic predispositions.

- Release of **pro-inflammatory mediators** (cytokines and chemokines)
- Recruitment of **immune cells** (mast cells)
- **Chronic inflammation**
- Dilated **blood vessels** and **redness** in the face, **swelling**, **itching** etc.

Pyroptosis

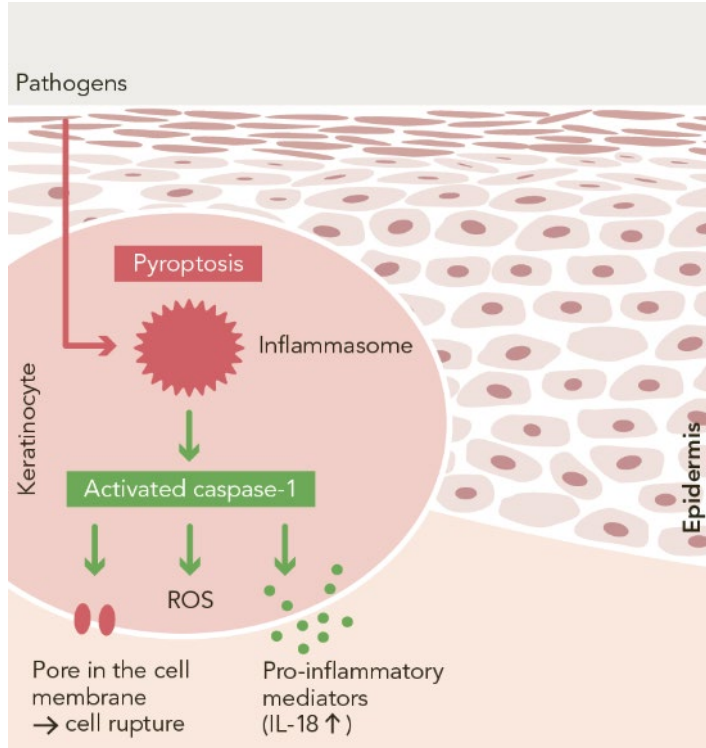
An Inflammatory Cell Death



- Recent research has unveiled a critical **link between rosacea and pyroptosis**, a highly **inflammatory** form of **programmed cell death** (1).
- The term is derived from the Greek words *pyro* (**fire**) and *ptosis* (**falling**).
- Unlike other forms of programmed cell death, pyroptosis ignites a **rapid** and **intense inflammatory response**.
- Role: **antimicrobial response**, effectively **eliminate infected cells**.

(1) Hu et al. (2023) *Experimental dermatology* 33(1): e14812.

Pyroptosis and the Role of Caspase-1

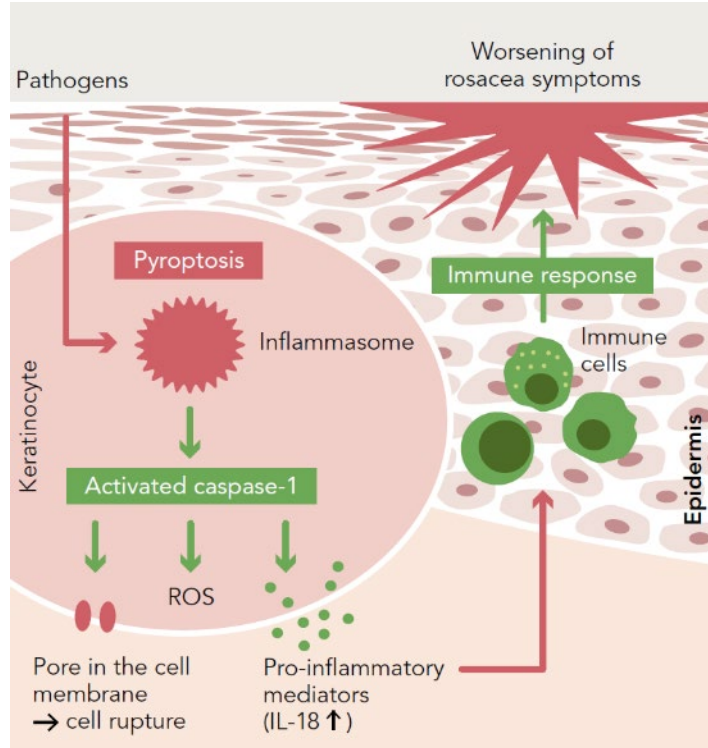


Pyroptosis involves the formation of a large protein complex known as the **inflammasome**.

Upon infection of the cell, the inflammasome activates **caspase-1**, the **key enzyme of pyroptosis**.

- Release of pro-inflammatory mediators such as **interleukin 18 (IL-18)**
- Generation of **reactive oxygen species (ROS)**
- Formation of **pores** in the cell membrane
- Loss of cell integrity, **cell rupture**
- **Release** of cell debris and pro-inflammatory mediators

Pyroptosis can Exacerbate Rosacea



In a next step, immune cells are ordered to the site → the extensive **immune response** harms nearby uninfected cells.

- Pyroptosis serves as a rapid defense mechanism, but it can also “**spread the fire**”.
- Leads to **chronic inflammation** and **tissue damage** if over-activated.
- Associated with various diseases, including autoinflammatory and cardiovascular diseases, as well as **rosacea**.



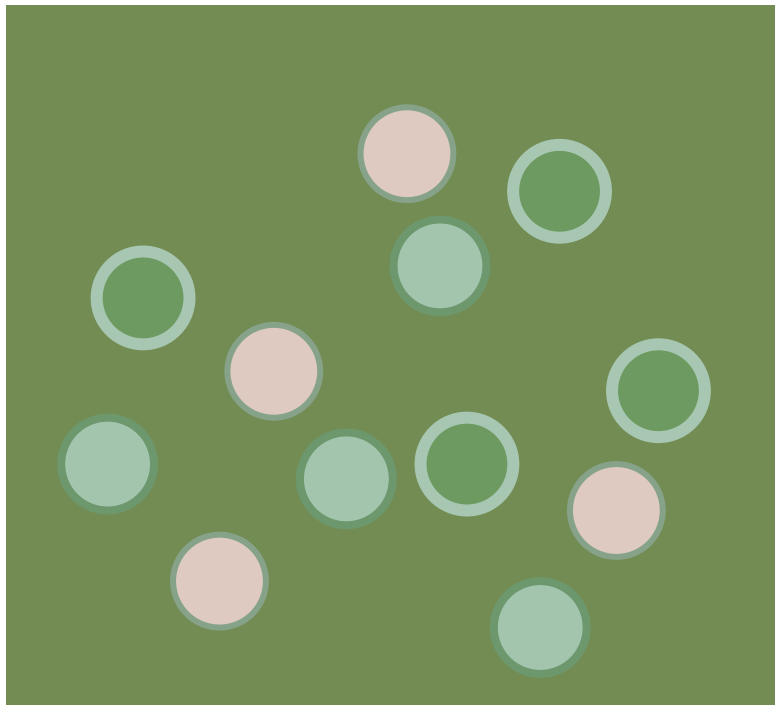
With **AmelioSense™**, Mibelle Biochemistry has found a new approach to effectively fight both, **pyroptosis** and **inflammation**.

Shepherd's Purse – a Herb with Anti-Pyrototic Properties



- Shepherd's purse is a member of the cruciferous family (*Brassicaceae*).
- The scientific name ***Capsella bursa-pastoris*** is made up of the Latin words *capsa* for “capsule,” *bursa* for “pocket,” and *pastor* for “shepherd.”
- Flat green fruits of the plant are shaped like the **pouches** of former **shepherds**.
- Shepherd's purse is a versatile **medicinal herb** with a long history in natural medicine.
- Primarily known for its **anti-inflammatory effects** and its ability to **stop bleeding** and **promote wound healing**.

AmelioSense™ - a Complex of Powerful Antioxidants and Shepherd's Purse Extract



AmelioSense™ is a complex that combines the calming properties of the **shepherd's purse extract** with a potent **liposomal antioxidant complex** consisting of:

- **carnosine**, a dipeptide that primarily fights skin glycation and oxidative stress
- **silymarin**, a flavonoid that is extracted from the milk thistle and is known for its anti-inflammatory properties
- **tocopherol**, a powerful oil-soluble antioxidant that protects against oxidative damage

The **liposomal** formulation improves the penetration of these ingredients into the skin.

→ Spray-granulated onto maltodextrin for better stability

→ **AmelioSense™**

AmelioSense™ Composition

Capsella Bursa-Pastoris Extract (dry)	0.8 %
Phospholipids	2.5 %
Carnosine	1.5 %
Tocopherol	0.04 %
Silymarin	0.01 %
Maltodextrin	89.45 %
Aqua (residual moisture)	~ 6 %



INCI (EU/PCPC) Declaration

Capsella Bursa-Pastoris Extract (and) Lecithin (and) Carnosine (and) Tocopherol (and) Silybum Marianum Fruit Extract (and) Maltodextrin (and) Aqua/Water

Recommended use level: 2 %



Inhibitory Effect on Pyroptosis Caspase-1, ROS and IL-18

Cell type: Normal human epidermal keratinocytes (NHEK)

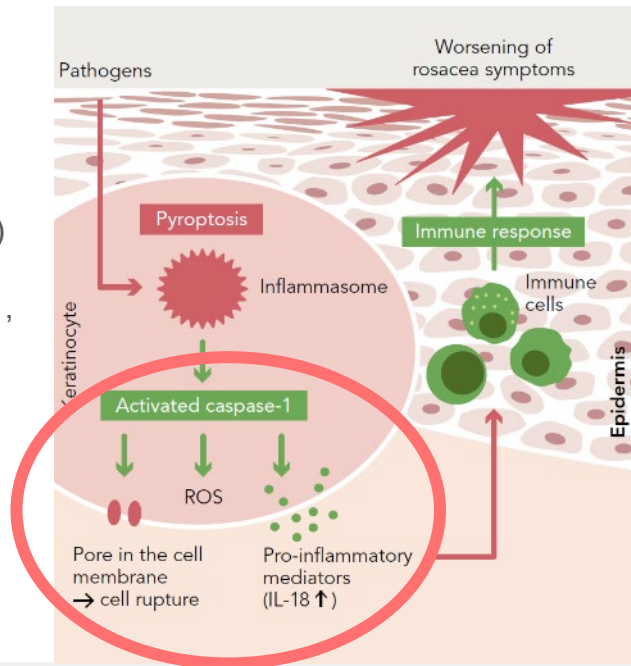
Test compound: 0.25 % shepherd's purse extract, control

Study design:

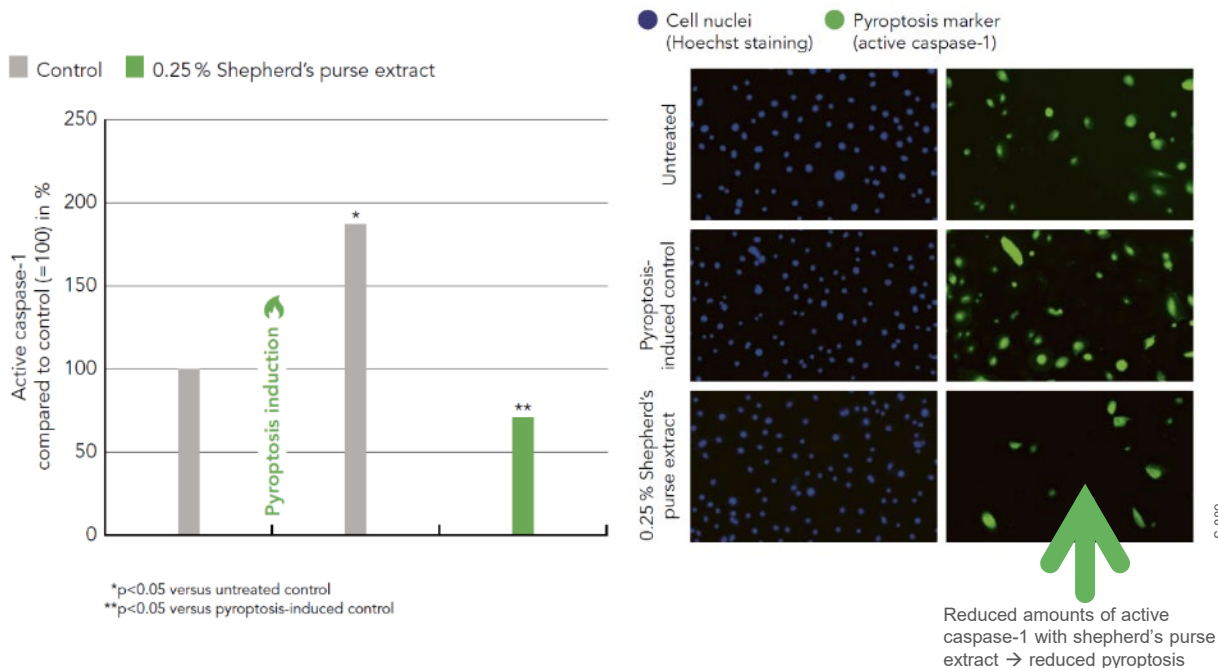
1. Pretreatment: 24 h with the test compounds or not (control)
2. Pyroptosis trigger: Replacement of the medium by a medium containing HMGB1 (high-mobility group protein B1, a pyroptosis trigger), either alone (trigger only), or together with the extract, for a further 24 h.

Parameters:

- Active caspase-1 (fluorescence microscopy)
- ROS release (CellROX green reagent, fluorescent staining)
- IL-18 gene expression



Inhibitory Effect on the Key Enzyme of Pyroptosis, Active Caspase-1

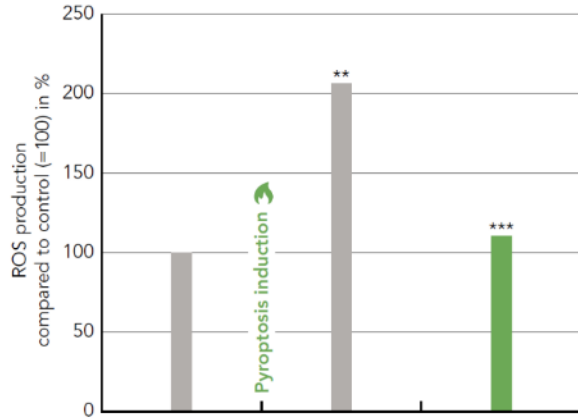


Shepherd's purse significantly reduced active caspase-1, the key enzyme in pyroptosis.

→ AmelioSense™ could **stop the spreading of the inflammation** and thus has the potential to **soothe** irritated, inflamed skin.

Reduction of ROS Release

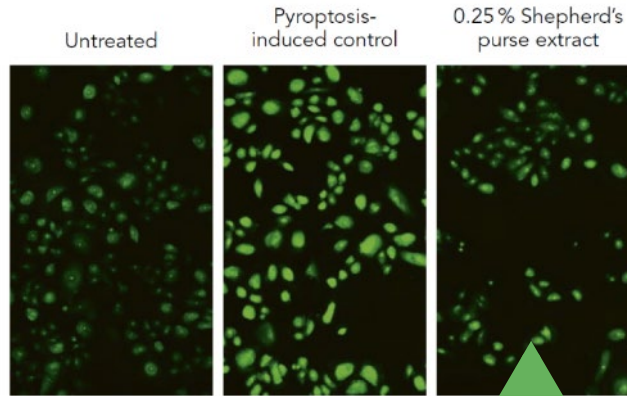
■ Control ■ 0.25 % Shepherd's purse extract



**p<0.01 versus untreated control

***p<0.01 versus pyroptosis-induced control

● ROS (CellROX)



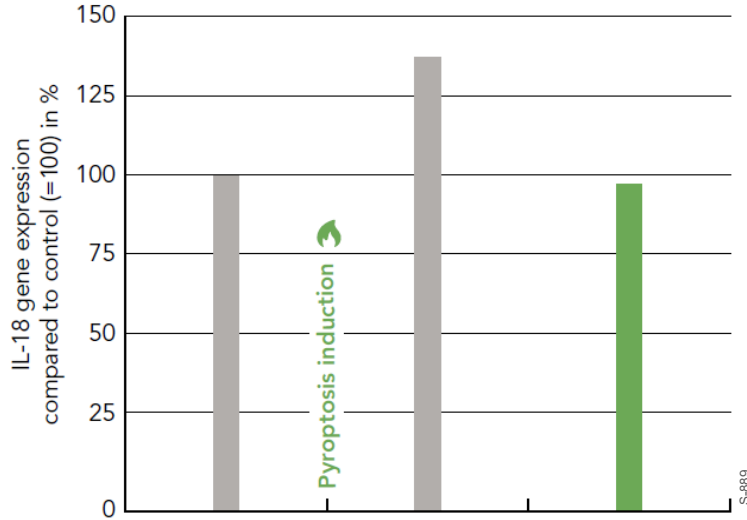
Reduced ROS release with shepherd's purse extract

The pyroptosis induction strongly increased ROS production, whereas shepherd's purse extract almost completely prevented their increase.

→ **AmelioSense™** reduces **oxidative stress** and thus could lead to a calmer skin.

Reduction in IL-18 Gene Expression

■ Control ■ 0.25 % Shepherd's purse extract



Pyroptosis induction increased the levels of the pro-inflammatory cytokine interleukin 18 (IL-18).

Pretreatment with shepherd's purse extract completely prevented the increase of IL-18 gene expression.



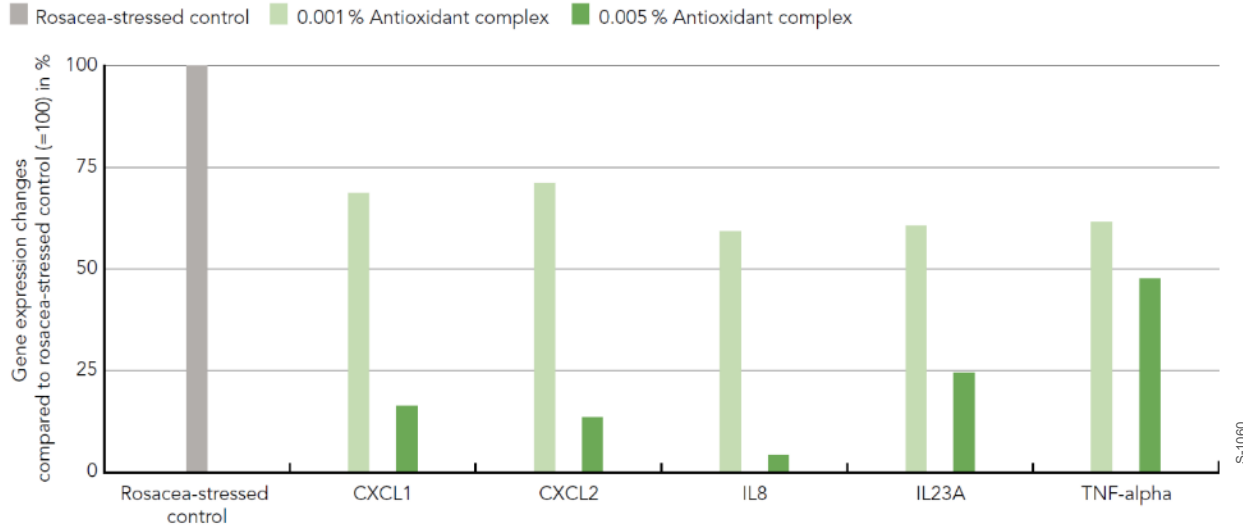
→ AmelioSense™ could strongly contribute to **combating inflammation** and **redness** in the skin.



Reduction of Inflammatory Markers

Cell type:	Normal human epidermal keratinocytes (NHEK)
Test compound:	0.001 % and 0.005 % of the antioxidant complex of AmelioSense™, control
Study design:	1. Pretreatment: 24 h with the test compounds. 2. Rosacea trigger: Replacement of the medium by a medium containing a combination of several rosacea-trigger factors (e.g. LL-37, calcitriol, IL-17) to induce an inflamed environment comparable to rosacea, either alone (= trigger only), or together with the extract for a further 48 h.
Parameters:	Gene expression (RT-qPCR)

Reduction of Inflammatory Markers



The antioxidant complex reduced gene expression of inflammatory markers, which are also known to be **crucial in rosacea** in a dose-dependent way.



The antioxidant complex has a potent **anti-inflammatory capacity**, which makes AmelioSense™ a powerful active in regard to **fighting skin inflammation and irritation**.



Reduction of Inflammation

Cell type: Normal human epidermal keratinocytes (NHEK)

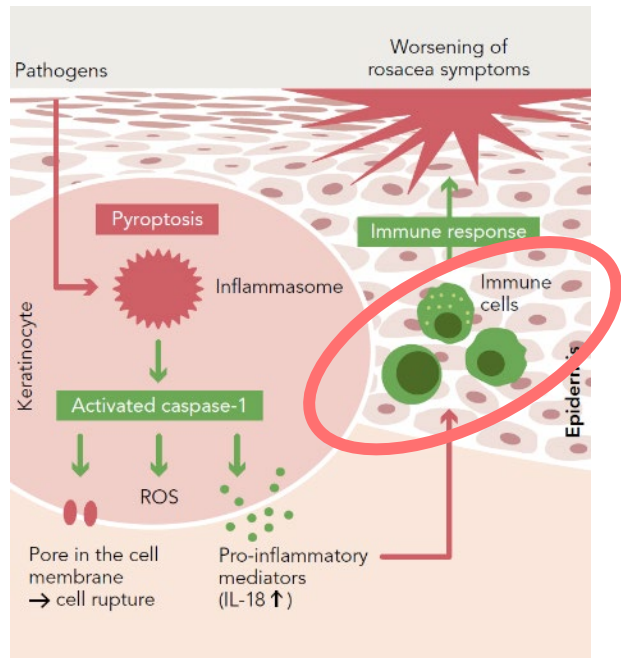
Test substances: 0.04 % shepherd's purse extract combined with antioxidant complex ($\triangleq$ AmelioSense™ suitable for *in vitro* testing), control

Study design:

1. Pretreatment with the AmelioSense™ combination or with ibuprofen (positive control) or left untreated (negative control) for 2 h.
2. Application of stress factor TNF-alpha to induce inflammation stress.

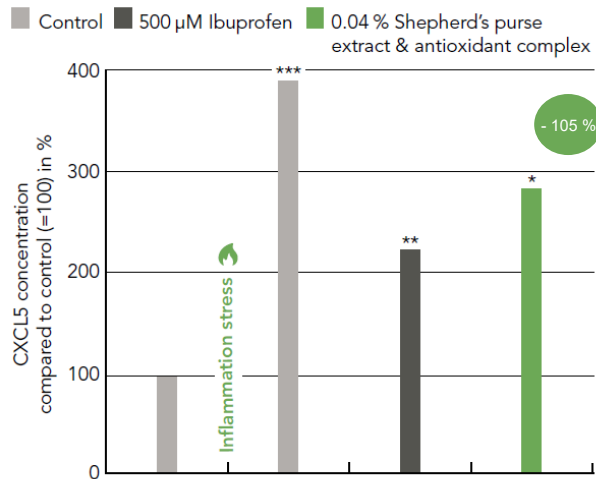
Parameters:

- Chemokine **CXCL5** (C-X-C Motif Chemokine Ligand 5) in the culture supernatants (ELISA)
 - Chemokine **MCP-1** (Monocyte Chemoattractant Protein-1) in the culture supernatants (ELISA)
- Both chemokines are involved in the **recruitment of immune cells** & thus **promotion of inflammation**



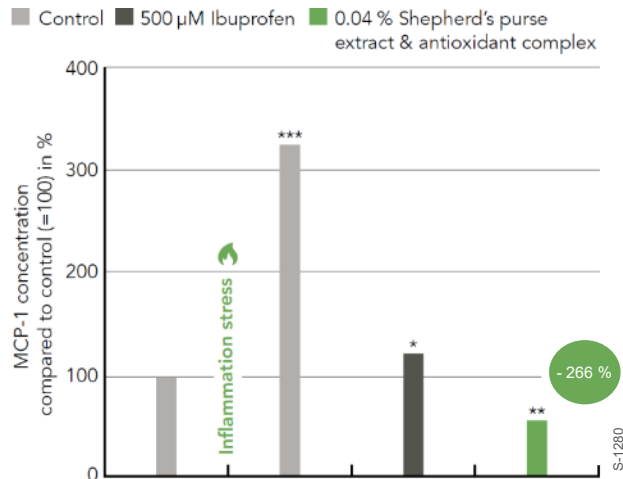
Reduction of Inflammation

CXCL5



*p<0.05 versus control and versus challenged control
 **p<0.01 versus control and versus challenged control
 ***p<0.001 versus control

MCP-1



*p<0.05 versus control and versus challenged control
 **p<0.01 versus control and versus challenged control
 ***p<0.001 versus control

Strong increase of the chemokines CXCL5 and MCP-1 after applying the inflammation stress (= TNF-alpha).

→ With the combined shepherd's purse extract & antioxidant complex, significant reduction of inflammation reaction by 105 % and 266 %, respectively (compared to stressed control).



The combined shepherd's purse extract and antioxidant complex ($\triangleq$ AmelioSense™) can effectively combat inflammation and thus contribute to skin soothing.

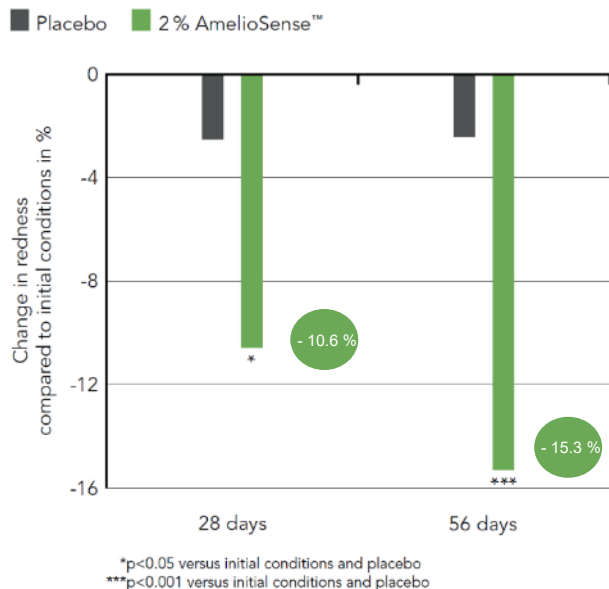


Visible Improvement of Skin Redness, Blood Flow and Blood Vessels



- Volunteers:** Double-blind, randomized, placebo-controlled clinical trial, 22 (m & f, 29 – 70 y) affected by redness, irritated skin and visible blood vessels in the cheek area
- Test product:** Cream with 2 % AmelioSense™, placebo
- Study design:** Application of the test product to one side of the face and the corresponding placebo cream to the other side twice a day
- Parameters:**
- Redness (high resolution standardized images, ColorFace®, Spectrophotometer® CM-700d)
 - Blood flow measured in tiny blood vessels under the epidermis (capillaries, Laser Doppler VMS-LDF2®)
 - Visible blood vessels & skin redness (close-up images, Dermatoscope® Delta20 Plus system mounted on a Canon EOS 450D reflex camera)
 - Questionnaire (self-evaluation on flushing and soothing effects)

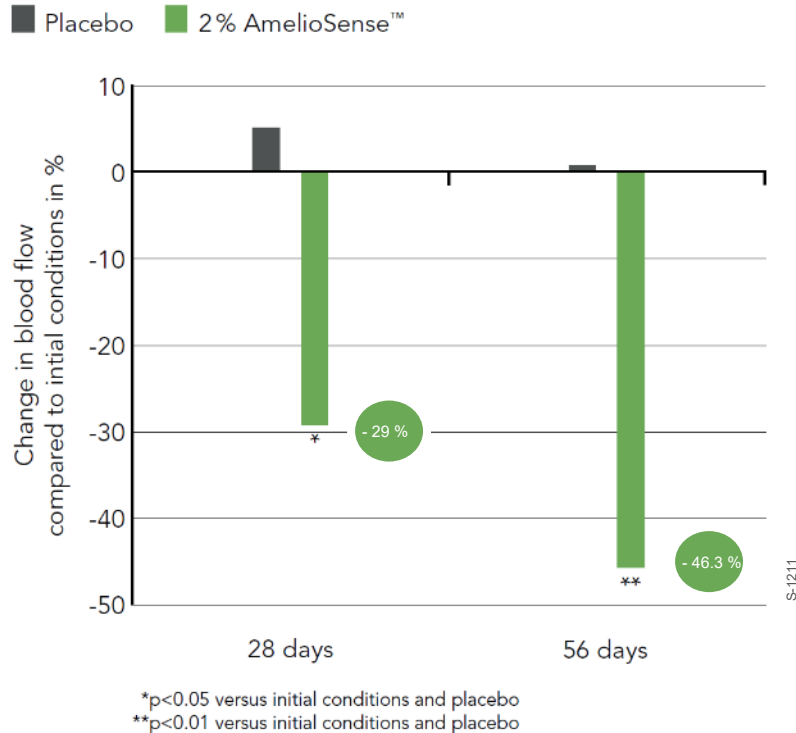
Visible Improvement of Skin Redness



AmelioSense™ significantly **reduced redness** after 28 days by 10.6 % and after 56 days and by 15.3 % compared to initial conditions.

→ AmelioSense™ significantly reduces visible skin redness, which can lead to improved quality of life.

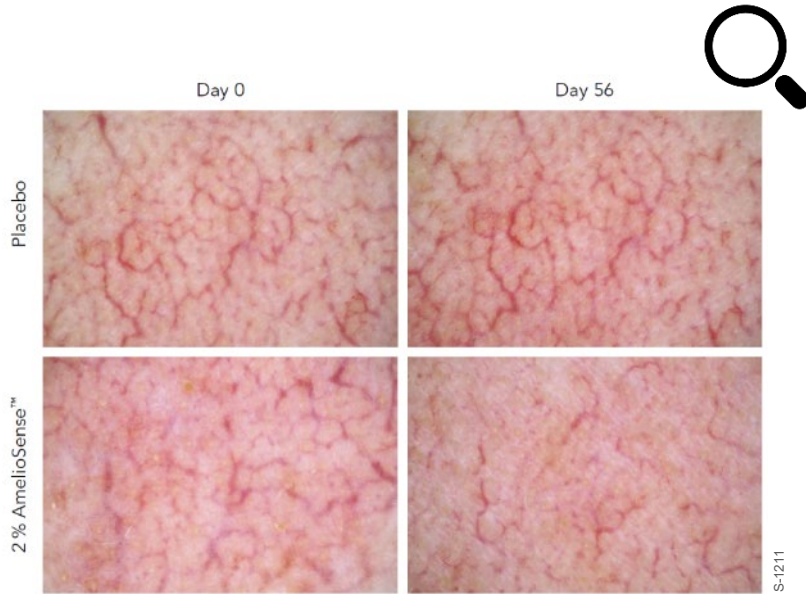
Reduction in Blood Flow



Blood flow was significantly reduced by 29 % after 28 days and by 46.3 % after 56 days compared to initial conditions.

→ The lower the blood flow, the less severe the skin flushes – **AmelioSense™** showed its ability to significantly **reduce blood flow** and thus **facial redness**.

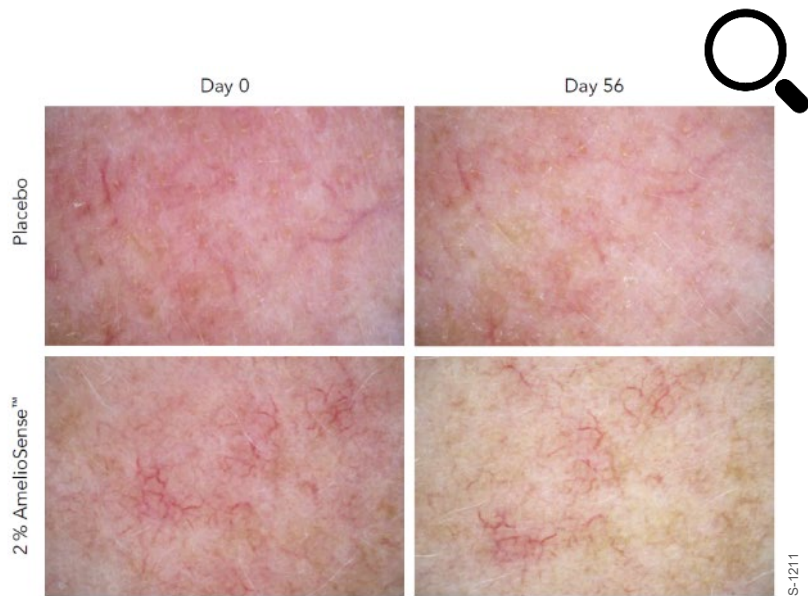
A Zoom-in to the Cheek Region: Reduction of Visible Blood Vessels



The zoomed images clearly showed the visible improvements in terms of **blood vessel appearance** after 56 days compared to initial conditions and compared to the placebo.

→ **AmelioSense™** helps to **improve the appearance** of the skin, providing a comprehensive solution for **redness** and **irritation of the skin**.

A Zoom-in to the Cheek Region: Reduction of Skin Redness



The close-up pictures showed that AmelioSense™ visibly **improved skin redness** after 56 days of treatment compared to initial conditions and compared to the placebo.

Questionnaire

Moreover, the volunteers self-evaluated AmelioSense™ with the help of a **questionnaire** at the end of the study

- 82 % of volunteers reported that their skin showed **reduced flushing effects**.
- 91 % of volunteers described that the product had a **soothing effect** on their skin.



AmelioSense™

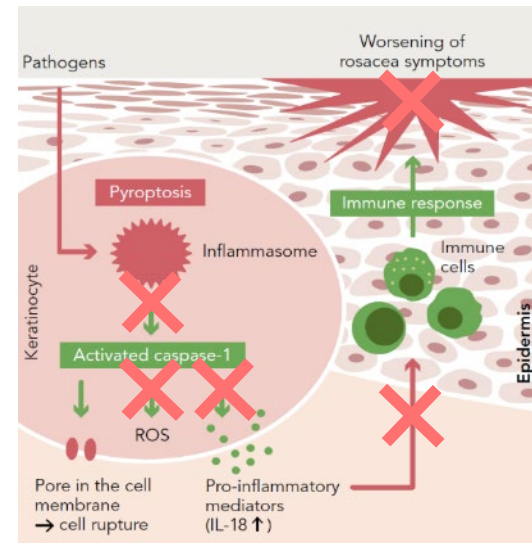
Summary Study Results



- Inhibitory effect on the key enzyme of pyroptosis, active caspase-1 (NHEK)
- Reduction of ROS release (NHEK)
- Reduction of IL-18 gene expression (NHEK)
- Reduction of gene expression of inflammatory markers relevant in rosacea (NHEK)
- Reduction of inflammation (chemokines CXCL5 and MCP-1, NHEK)



- Significant improvement of skin redness
- Reduction in blood flow
- Zoom-in: reduction of visible blood vessels
- Zoom-in: reduction of redness
- Questionnaire: reduction of flushes and a soothing effect



✗ Proven activities of AmelioSense™

AmelioSense™



- Reduces the appearance of visible blood vessels on the cheeks
- Decreases facial redness and blood flow
- Alleviates irritation
- Promotes a calm and balanced skin tone

AmelioSense™ Applications



- Soothing anti-redness serum for delicate skin
- Redness relief soothing cream
- Ultra calming anti-redness mask
- Instant relief barrier strengthening serum
- Redness relief daily cleansing milk

AmelioSense™

Marketing Benefits



- Clearly visible effects
- Gentle soothing ingredient to treat irritated and flushed skin
- Targets pyroptosis, a new mechanism that underlies inflammation and rosacea
- Shortlisted product for in-cosmetics Global Innovation Zone Best Ingredient Award 2025 - Active Ingredient



Marketing Tools



Prototype Formulation

Soothing Anti-Redness Serum

→ PDF available, prototype samples, [blog post](#)



Movies

[Teaser Movie](#)

[Product Movie](#)

Blog

[Rosacea and Pyroptosis - an inflammatory alliance](#)

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